

ORIGINAL RESEARCH ARTICLE

Automated Echocardiographic Detection of Congenital Heart Disease Using Artificial Intelligence

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BACKGROUND: Delayed or missed diagnosis of congenital heart disease (CHD) contributes to excess pediatric mortality worldwide. Echocardiography (echo) is central to diagnosing and triaging CHD, yet expert interpretation remains a scarce and maldistributed global resource. Artificial intelligence offers the potential to democratize diagnostics and to extend expert-level interpretation beyond large academic centers, but its application in CHD remains underexplored.

METHODS: We developed EchoFocus-CHD, an artificial intelligence-enabled model for automated detection of 12 critical and 8 noncritical CHD lesions, individually and as composites. The composite critical CHD outcome was the primary end point. The model expands on a multitask, view-agnostic architecture (PanEcho) with a transformer encoder to improve focus on relevant echo views. The model was internally trained (80%) and tested (20%) on the first echo per patient from Boston Children's Hospital, with further evaluation on a referral cohort of echo studies performed at external US and international centers.

RESULTS: The internal and referral cohorts included 3.4 million videos from 54 727 echos (median age at echo, 7.1 years [interquartile range, 0.2–15.0 years]; 5.8% critical CHD, 23.6% noncritical CHD) and 167 484 videos from 3356 echos (median age at echo, 2.5 years [interquartile range, 0.3–9.4 years]; 29.4% critical CHD, 45.6% noncritical CHD), respectively. EchoFocus-CHD showed excellent internal ability to detect the composite critical CHD outcome (area under the receiver-operating curve [AUROC], 0.94; positive likelihood ratio, 7.50; negative likelihood ratio, 0.14) and individual critical lesions (AUROC, 0.83–1.00), as well as composite noncritical CHD (AUROC, 0.90; positive likelihood ratio, 5.00; negative likelihood ratio, 0.23) and individual noncritical lesions (AUROC, 0.70–0.96). Performance declined during evaluation on the referral cohort to detect critical CHD (AUROC, 0.77), coinciding with greater expert disagreement on referral cases ($\kappa=0.72$ versus 0.82 for internal cases). Explainability analyses demonstrated that the model prioritized the same clinically relevant views (parasternal long axis, parasternal short axis, subxiphoid long axis, apical) across internal and referral cohorts, whereas uniform manifold approximation and projection analysis revealed a domain shift between cohorts. Retraining on all available US patients attenuated domain shift effects, improving international critical CHD detection (AUROC, 0.87) and calibration.

CONCLUSIONS: EchoFocus-CHD shows promise for automated CHD detection to advance equitable global cardiovascular care and highlights the need to address domain shift and to establish external validation before real-world deployment.

Key Words: artificial intelligence ■ cardiology ■ echocardiography ■ heart defects, congenital ■ pediatrics

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Clinical Perspective

What Is New?

- EchoFocus-CHD is a view-agnostic, multitask artificial intelligence model capable of automatically detecting congenital heart disease (CHD) from full echocardiogram studies.
- To the best of our knowledge, this study represents the largest pediatric echocardiogram dataset to date (≈ 3.6 million videos from $\approx 58\,000$ studies across 58 countries and 6 continents), capturing widespread diversity in demographics and CHD lesions.
- EchoFocus-CHD demonstrates excellent internal performance (area under the receiver-operating curve, 0.94 for composite critical CHD) and shows domain shift on a referral cohort that can be partially mitigated with the use of a diverse training cohort, highlighting the need for external validation and practical strategies and challenges for global artificial intelligence deployment.

What Are the Clinical Implications?

- EchoFocus-CHD may offer a pathway to equitable artificial intelligence–assisted pediatric cardiology care by extending limited specialist expertise worldwide.
- Automated critical CHD detection can help prioritize high-risk studies for expert review, potentially reducing delays in diagnosis and treatment, especially in resource-limited settings.
- Model explainability analyses consistently identified clinically relevant echocardiographic views, which may support clinician trust in artificial intelligence–enabled triage and decision-making.

Nonstandard Abbreviations and Acronyms

AI	artificial intelligence
AUROC	area under the receiver-operating curve
BCH	Boston Children's Hospital
CHD	congenital heart disease
echo	echocardiography
LR	likelihood ratio
UMAP	uniform manifold approximation and projection

Congenital heart disease (CHD) affects ≈ 1 in 100 live births, affecting >12 million individuals worldwide.^{1,2} Nearly 25% of CHD cases are critical, often requiring urgent intervention in the neonatal period to prevent cardiovascular collapse and death.³ Unfortunately, CHD is frequently diagnosed late in both low-resource⁴ and high-resource⁵ countries, reflecting a persistent diagnostic gap. This challenge is particularly

severe in low- and middle-income countries where the burden of disease is greatest⁶ and access to diagnostics and congenital care is limited,^{6,7} highlighting the global imperative for timely and effective CHD detection and triage.

Echocardiography (echo) is the cornerstone of pediatric cardiology and CHD diagnosis, providing noninvasive, real-time assessment of cardiac anatomy and function without radiation. Pediatric echo interpretation is technically challenging: It requires the interpretation of complex, heterogeneous lesions in small hearts and is often complicated by motion artifacts and variable image quality. These challenges are compounded by a global shortage of pediatric cardiologists and specialized imaging experts,^{6–8} creating a critical bottleneck for timely and accurate diagnoses.

Artificial intelligence (AI) has shown promise to address diagnostic gaps in adult echo. For example, AI-echo models can reliably automate measurements,^{9–12} assess heart muscle and valve function,¹³ or even provide a comprehensive echo evaluation.¹⁴ In contrast, transthoracic AI-echo for pediatric cardiology remains nascent, with prior work largely limited to view classification,¹⁵ isolated measurement tasks,^{16,17} or detection of specific findings (eg, patent ductus arteriosus)¹⁸ rather than comprehensive structural screening.¹⁹

To address this technological gap, we developed EchoFocus-CHD, a multitask, view-agnostic AI-echo model designed to automatically detect a broad spectrum of critical and noncritical CHD lesions. To assess generalizability and performance under real-world conditions, we evaluated the model using echos from 58 countries across 6 continents with the goal of enabling scalable CHD triage and prioritization in resource-limited settings.

METHODS

This study is reported in accordance with the TRUE-AIM (Transparent Reporting and Evaluation of Artificial Intelligence Models) report card.²⁰

Patient Population and Patient Assignment

Patient data and echos were obtained from Boston Children's Hospital (BCH) between July 2015 and July 2025. To exclude echos containing insufficient video data for reliable CHD assessment, only transthoracic echos with ≥ 10 DICOM files were included. Given our objective to identify previously unknown or unverified CHD, only the first echo per patient was included. These criteria defined the main study cohort.

The main cohort was subsequently partitioned into internal studies (performed at BCH, Brigham and Women's Hospital nursery/neonatal intensive care unit, Beth Israel nursery/neonatal intensive care unit, or affiliated BCH satellite clinics) and referral studies (echos performed at outside institutions and subsequently read by BCH expert cardiac imagers for diagnostic assistance or second opinions). The referral cohort

was further subdivided into US and international patients. International patients were defined as having non-US home addresses. Within the internal cohort, patients were randomly assigned in an 80:20 ratio to the development and testing cohorts.

Definition of Outcomes

Diagnostic labels for each echo were derived from the Fyler coding system, a detailed, decades-old, well-established anatomic classification system used at BCH and specifically designed for CHD.²¹ For every echo, expert interpreting cardiac imagers (with subspecialty training in noninvasive pediatric cardiac imaging) assign Fyler codes for internal and referral studies that capture both major and minor structural cardiac lesions with high anatomic granularity. In a randomly sampled subset of 100 echo reports, Fyler codes demonstrated high agreement with manual annotation (97% accuracy for atrial septal defects and 99%–100% for all other CHD labels).

Outcomes of interest included critical and noncritical CHD lesions, predicted individually and as composites (Table S1). CHD lesions were considered critical if surgical or catheter-based intervention is typically required within the first year of life. A CHD lesion was considered as noncritical if it is typically managed conservatively or with intervention delayed beyond infancy. The composite critical CHD outcome was the primary end point. Outcome labels were not mutually exclusive (ie, a patient can have tetralogy of Fallot and an atrial septal defect).

Data Retrieval and Preprocessing

All echo studies were retrieved from the institutional picture archiving and communication system. All echos underwent preprocessing analogous to that described in the PanEcho framework.⁹

Pixel data from 2-dimensional echo videos were first extracted from DICOM files. All still frames images (inclusive of images containing measurements or annotations) were excluded. All videos then underwent comprehensive deidentification. Specifically, each frame was binarized with the use of a fixed threshold, and all pixels outside the convex hull of the largest detected contour were masked. Videos were subsequently cropped to the central image content in a temporally consistent manner, downsampled to a resolution of 256×256 pixels with bicubic interpolation, and further deidentified by masking peripheral regions containing protected health information.⁹ Videos were then augmented (see the Model Training section for details) and centrally cropped to 224×224 pixels.

EchoFocus-CHD Model Architecture

The EchoFocus-CHD architecture takes a set of echo videos from a single study as input and produces multiple task-specific predictions of CHD classifications. The architecture extends PanEcho⁹ by adapting the final layers of the network with additional transformer layers to allow attention²² to operate over video clip embeddings (Figure 1B). Analogous to how a human expert interprets an echo, the attention mechanism enables the model to selectively weight diagnostically informative videos, enhancing the representation of relevant structural and functional features for CHD classification.

Echo videos are first separated into 16 random sets of 16 sequential frames (called clips); each frame (image) is individually processed with a 2-dimensional convolutional neural network (ConvNeXt-T,²³ pretrained on ImageNet) to produce image embeddings. These image embeddings are stacked sequentially and fed into a temporal transformer, consisting of 4 layers with 8 attention heads. This process mimics the use of transformers for interpreting natural language sentences; in this setting, the image embeddings are like word “tokens,” and the clips are treated as “sentences.” To capture the temporal information of the frames, a standard positional encoding is added to the image tokens. For each clip in the echo study, the output of the temporal transformer is aggregated with mean pooling to produce a clip-level embedding, represented as a 768-dimensional vector.

EchoFocus-CHD then departs from the PanEcho architecture⁹ by introducing a study-level transformer encoder that operates across all (number of videos×16) clip-level embeddings to generate a single study-level embedding. This transformer encoder leverages self-attention to learn additional dependencies between videos in the study before moving to task prediction. The resulting study-level embedding is then passed through fully connected layers to generate task-specific outputs of CHD classification labels.

Model Training

The internal BCH model development cohort was randomly partitioned at the patient-level into training (80%) and validation (20%) sets. The model was trained using the training set, with the validation set used exclusively for model selection. During training, pretrained PanEcho model weights were frozen and used to generate video-level embeddings, allowing optimization to focus on learning the parameters of the study-level transformer encoder and the fully connected, task-specific output layers.

Training was performed with the AdamW optimizer²⁴ with a weight decay of 0.01 and a scheduled learning rate that decreased on plateaus in validation loss. Training was terminated after 10 consecutive epochs without improvement in validation loss.

Several strategies were used to improve training robustness. Consistent with the PanEcho approach,⁹ we used several image augmentation techniques (cropping, rotation, and flipping) to improve robustness to imaging noise. On layers following PanEcho, dropout²⁵ was applied during training at a rate of 0.2 with an additional clip-level dropout at 0.5 to enhance robustness to missing video clips.

For hyperparameter tuning, we varied the depth of the study-level transformer encoder (1, 5, 10, and 20 layers), the learning rate (0.0001–0.01), and the effective batch size (32–128). The final model hyperparameters were selected by minimizing loss across tasks on the held-out validation set.

Model Performance Evaluation and Statistical Analyses

Model discrimination was assessed with the area under the receiver-operating characteristic curve (AUROC). Additional clinically relevant performance metrics included sensitivity, specificity, positive and negative predictive values, positive and

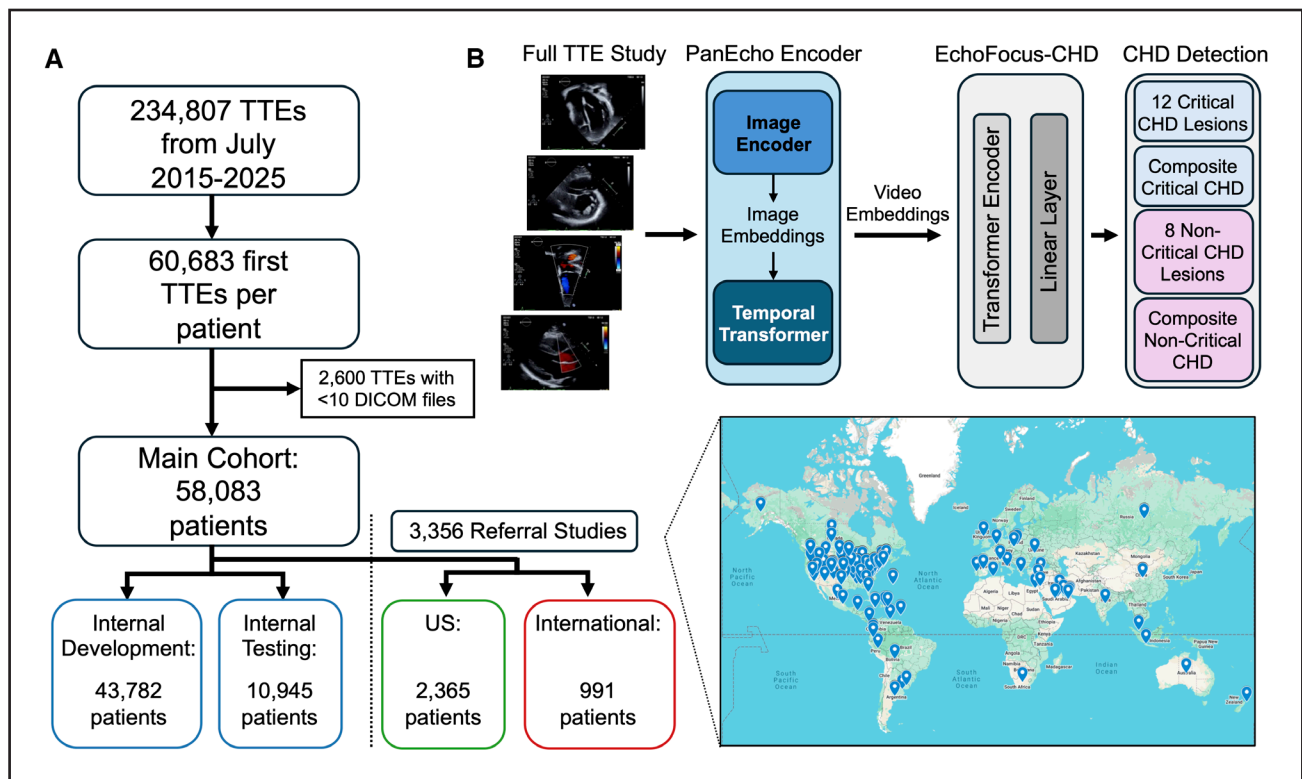


Figure 1. Schematic of study design and model architecture.

A, Schematic of training and testing design. STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) diagram showing initial patient selection and filtering at each data processing stage (with primary outcome rates shown). **Inset**, Pins of origin countries for referral studies. **B**, Schematic of EchoFocus-CHD architecture and classification targets. TTE indicates transthoracic echocardiogram.

negative likelihood ratios (LRs), and lift. These metrics were computed with decision thresholds that maximize the Youden index, derived from the validation set. The resulting threshold was then fixed and applied uniformly across all internal and referral test cohorts. CIs for performance metrics were estimated from 1,000 bootstrap samples.

Model Calibration Analysis

Model calibration was assessed through calibration plots and scaled Brier scores. Scaled Brier scores account for differences in outcome prevalence across cohorts and provides an interpretable metric ranging from 0 (no improvement over baseline) to 1 (perfect prediction).

Sensitivity and Subgroup Analyses

We evaluated the robustness of the model for detecting exclusively unrepaired CHD through a sensitivity analysis that excluded echos from patients with prior cardiac interventions (ie, catheterization or surgery), as determined by Fyler codes. To assess sensitivity to outcome labeling, we compared model performance when using structured Fyler code labels and labels automatically extracted from echo report text by an internal instance of GPT-4o-mini (OpenAI, San Francisco, CA).

Subgroup analyses were performed on the test cohorts stratified by age and number of echo videos per study. Age groupings were adapted from prior work²⁶ and defined as <1 year of age (infant); 1 to 3, 3 to 8, 8 to 12, and 12 to 18 years of age; and >18 years of age. Echo videos per study groupings

were defined as <25, 26 to 50, 51 to 75, 76 to 100, and >100. Model discrimination within each age subgroup was assessed with the AUROC.

Model Adjudication

Four expert cardiac imagers characterized model errors through an adjudication process. For both internal and referral studies, 2 experts each independently reviewed 25 false-positive and 25 false-negative infant echos. These echos were selected by simple random sampling; no stratification or selection based on predicted probability or clinical characteristics was performed.

Adjudicators reviewed the full echo study and were blinded to patient name, echo report, model predictions, and each other's assessments. For each echo, adjudicators were asked to classify the study into 1 of 4 categories: (1) critical CHD, (2) noncritical CHD, (3) indeterminate because of inadequate image quality, or (4) indeterminate because of evolving physiology requiring follow-up such as suspected coarctation of the aorta in the presence of a ductus arteriosus. Adjudication outcomes between internal and referral cohorts were compared with the Fisher exact test.

For the purposes of evaluating agreement in a triage context, we calculated the Cohen κ when grouping indeterminate studies with noncritical studies to yield a binary critical versus noncritical/indeterminate classification. A Cohen κ value of 1 indicates perfect agreement; a value of 0 indicates agreement equivalent to chance; and values <0 indicate agreement worse than chance.

Model Explainability

To evaluate whether accurate model predictions were supported by clinically appropriate echo views, the explainability analysis was restricted to studies with the smallest prediction errors. Examining cases in which the predictions of the model closely matched the reference diagnosis provides the clearest view of the diagnostic pathway underlying the decisions of the model.

An integrated gradients-based explainability analysis was performed for 1 left-sided lesion (hypoplastic left heart syndrome) and 1 right-sided lesion (tetralogy of Fallot). For each lesion, we selected 25 internal and 25 referral echo studies with positive cases and the smallest prediction errors. For each echo study, integrated gradients were applied to quantify the contribution of individual video clips to the predicted output of the model.

The 10 most highly weighted video clips per study were identified and subsequently reviewed by an expert cardiac imager, who recorded (1) which unique echo views the model prioritized and (2) whether the 5 or 10 highest prioritized video clips were sufficient to detect the lesion of interest.

To further assess model sensitivity to specific echo views, we also conducted ablation experiments focusing on 4 views commonly obtained by sonographers and identified as high attention: parasternal long axis, parasternal short axis, subxiphoid long axis, and apical. Two cardiologists independently reviewed 25 positive and 25 negative internal echo studies for the same left-sided (hypoplastic left heart syndrome) and right-sided (tetralogy of Fallot) lesions to assign views to each clip. Ablation experiments included random exclusion of 25% of clips, exclusion of individual views, exclusion of all 4 views, and analysis using only the 4 selected views.

Embedding Visualization for Domain Shift Assessment

To explore potential domain shift²⁷ (ie, differences in training versus deployment echo imaging conditions that can degrade performance) between internal and referral echo studies, we applied unsupervised uniform manifold approximation and projection (UMAP) on high-dimensional embeddings produced by the EchoFocus-CHD study-level transformer encoder. We applied UMAP using 15 neighbors and the cosine distance metric. The resulting space was visualized and qualitatively compared between internal and referral cohorts to assess overlap and separation that might indicate domain shift related to differences in acquisition setting, patient population, or imaging protocols.

To move beyond qualitative assessment, we additionally performed quantitative analyses to formally evaluate distributional differences between cohorts. First, we conducted a clustering analysis (agglomerative clustering with 3 clusters) within the embedding space and assessed cohort compositions of each cluster. Second, we calculated the maximum mean discrepancy between internal and referral embeddings.

Data Availability and Software

The model and source code are available from <https://echofocus.org> for noncommercial, academic-only purposes to accelerate research on AI-echo in pediatric cardiology. All analyses

were implemented in Python (version 3.12). Requests for BCH data and related materials will be internally reviewed to clarify whether the request is subject to intellectual property or confidentiality constraints. Shareable data and materials will be released under a material transfer agreement for noncommercial research purposes. Use of BCH data was approved by its Institutional Review Board.

RESULTS

Patient Population Characteristics

From the 234 807 transthoracic echos at BCH meeting the inclusion criteria, 60 683 were first time studies per patient. After the exclusion of echos with <10 DICOM files per study ($n=2600$), 58 083 remained, forming the main cohort (Figure 1A). Of those, 54 727 were internal studies and 3356 referral studies were sent from outside centers: 2365 from patients across the United States and 991 from international patients. International patients resided in 58 countries spanning 6 continents: North America, South America, Europe, Asia, Africa, and Australia.

As shown in the Table, there were numerous differences between the internal and referral cohorts. There were 2.6 million, 0.8 million, and 0.2 million videos within the internal development, internal testing, and referral cohorts, respectively (Table). The internal studies had more videos per study (median, 75) compared with referral studies (median, 46). Internal studies were performed at an older age (median age at echo, 7.1 years [interquartile range, 0.2–15.0 years]) compared with referral studies (median age at echo, 2.5 years [interquartile range, 0.3–9.4 years]). There was a substantially higher prevalence of CHD in the referral cohort (29.4% critical CHD, 45.6% noncritical CHD) compared with the internal cohort (5.8% critical CHD, 23.6% noncritical CHD). The Table provides details of prevalence for individual lesions within each cohort.

EchoFocus-CHD Model Performance

Model performance metrics of EchoFocus-CHD for individual critical CHD lesions during internal and referral cohort testing are shown in Figure 2 and Tables S1 through S4. During internal testing, performance was excellent for a majority of lesions: AUROC of 0.97 for Ebstein anomaly; AUROC ≥ 0.99 for single-ventricle lesions; AUROC ≥ 0.97 for conotruncal lesions; AUROC of 0.96 for atrioventricular canal defects and pulmonary atresia; AUROC of 0.90 for coarctation of the aorta; and AUROC of 0.83 for total anomalous pulmonary venous connection. In comparison, there was a reduction in performance for the overall referral cohort across all individual critical CHD lesions, with the AUROC ranging from 0.70 to 0.85 (Figure 2).

For individual noncritical CHD lesions, internal performance ranged from an AUROC of 0.70 (anomalous

Table. Baseline Characteristics of Internal and Referral Cohorts

	Internal cohort		Referral cohort		
	Development	Testing	US	International	Overall
Patients, n	43782	10945	2365	991	3356
TTEs, n	43782	10945	2365	991	3356
Videos, n	2617348	818487	119152	48332	167484
Videos per study, n	75 (63–87)	75 (63–88)	45 (31–65)	47 (30–65)	46 (31–65)
Age at TTE, y	7.11 (0.25–14.96)	6.96 (0.20–14.99)	2.17 (0.34–8.96)	3.3 (0.33–10.40)	2.46 (0.33–9.36)
Sex (male), n (%)	23291 (53.20)	5752 (52.55)	1295 (54.76)	529 (53.38)	1824 (54.35)
Composite critical CHD, n (%)	2525 (5.77)	628 (5.74)	810 (34.25)	177 (17.86)	987 (29.41)
ALCAPA	7 (0.02)	2 (0.02)	3 (0.13)	0 (0.00)	3 (0.09)
AP window	13 (0.03)	5 (0.05)	7 (0.30)	0 (0.00)	7 (0.21)
DORV	163 (0.37)	35 (0.32)	116 (4.90)	30 (3.03)	146 (4.35)
D-loop TGA	235 (0.54)	62 (0.57)	55 (2.33)	17 (1.72)	72 (2.15)
Ebstein	83 (0.19)	25 (0.23)	54 (2.28)	17 (1.72)	71 (2.12)
HLHS	194 (0.44)	42 (0.38)	70 (2.96)	24 (2.42)	94 (2.80)
IAA	33 (0.08)	16 (0.15)	10 (0.42)	2 (0.20)	12 (0.36)
Tricuspid atresia	82 (0.19)	13 (0.12)	11 (0.47)	7 (0.71)	18 (0.54)
Truncus arteriosus	48 (0.11)	13 (0.12)	17 (0.72)	4 (0.40)	21 (0.63)
SV disease	330 (0.75)	70 (0.64)	113 (4.78)	41 (4.14)	154 (4.59)
Tetralogy of Fallot	515 (1.18)	104 (0.95)	86 (3.64)	16 (1.61)	102 (3.04)
AVCD	406 (0.93)	109 (1.00)	254 (10.74)	35 (3.53)	289 (8.61)
CoA	806 (1.84)	203 (1.85)	131 (5.54)	21 (2.12)	152 (4.53)
TAPVC	71 (0.16)	13 (0.12)	20 (0.85)	8 (0.81)	28 (0.83)
Critical AS	10 (0.02)	2 (0.02)	4 (0.17)	1 (0.10)	5 (0.15)
Critical PS	26 (0.06)	8 (0.07)	2 (0.08)	0 (0.00)	2 (0.06)
Pulmonary atresia	244 (0.56)	55 (0.50)	85 (3.59)	26 (2.62)	111 (3.31)
Composite noncritical CHD, n (%)	10277 (23.47)	2619 (23.93)	1232 (52.09)	298 (30.07)	1530 (45.59)
ASD	3037 (6.94)	771 (7.04)	441 (18.65%)	95 (9.59)	536 (15.97)
Anomalous coronaries	302 (0.69)	69 (0.63)	49 (2.07%)	6 (0.61%)	55 (1.64)
BAV	1212 (2.77)	287 (2.62)	212 (8.96)	35 (3.53)	247 (7.36)
Cor triatriatum	23 (0.05)	4 (0.04)	5 (0.21)	1 (0.10)	6 (0.18)
Double aortic arch	82 (0.19)	17 (0.16)	10 (0.42)	0 (0.00)	10 (0.30)
Left PA sling	15 (0.03)	3 (0.03)	4 (0.17)	1 (0.10)	5 (0.15)
LSVC	574 (1.31)	141 (1.29)	85 (3.59)	13 (1.31)	98 (2.92)
L-loop TGA	61 (0.14)	19 (0.17)	68 (2.88)	27 (2.72)	95 (2.83)
PAPVC	266 (0.61)	63 (0.58)	40 (1.69)	10 (1.01)	50 (1.49)
PDA	4521 (10.33)	1238 (11.31)	198 (8.37)	94 (9.49)	292 (8.70)
Right aortic arch	478 (1.09)	103 (0.9%)	67 (2.83)	9 (0.91)	76 (2.26)
Vascular ring	125 (0.29)	31 (0.28)	14 (0.59)	0 (0.00)	14 (0.42)
VSD	3177 (7.26)	794 (7.25)	461 (19.49)	95 (9.59)	556 (16.57)

Data presented as frequency (percentage) and median (interquartile range).

ALCAPA indicates anomalous left coronary artery from the pulmonary artery; AP, aortopulmonary; AS, aortic stenosis; ASD, atrial septal defect; AVCD, atrioventricular canal defect; BAV, bicuspid aortic valve; CHD, congenital heart disease; CoA, coarctation of the aorta; DORV, double-outlet right ventricle; HLHS, hypoplastic left heart syndrome; IAA, interrupted aortic arch; LSVC, left superior vena cava; PA, pulmonary artery; PAPVC, partial anomalous pulmonary venous connection; PDA, patent ductus arteriosus; PS, pulmonary stenosis; SV, single ventricle; TAPVC, total anomalous pulmonary venous connection; TGA, transposition of the great arteries; TTE, transthoracic echocardiogram; and VSD, ventricular septal defect.

coronaries) to 0.96 (ductus arteriosus). For atrial and ventricular septal wall defects, the AUROC was 0.87 and 0.91, respectively (Table S2). In the referral cohort,

performance also declined for noncritical CHD lesions. For example, referral AUROC decreased to 0.80 for patent ductus arteriosus, 0.74 for atrial septal defect, and

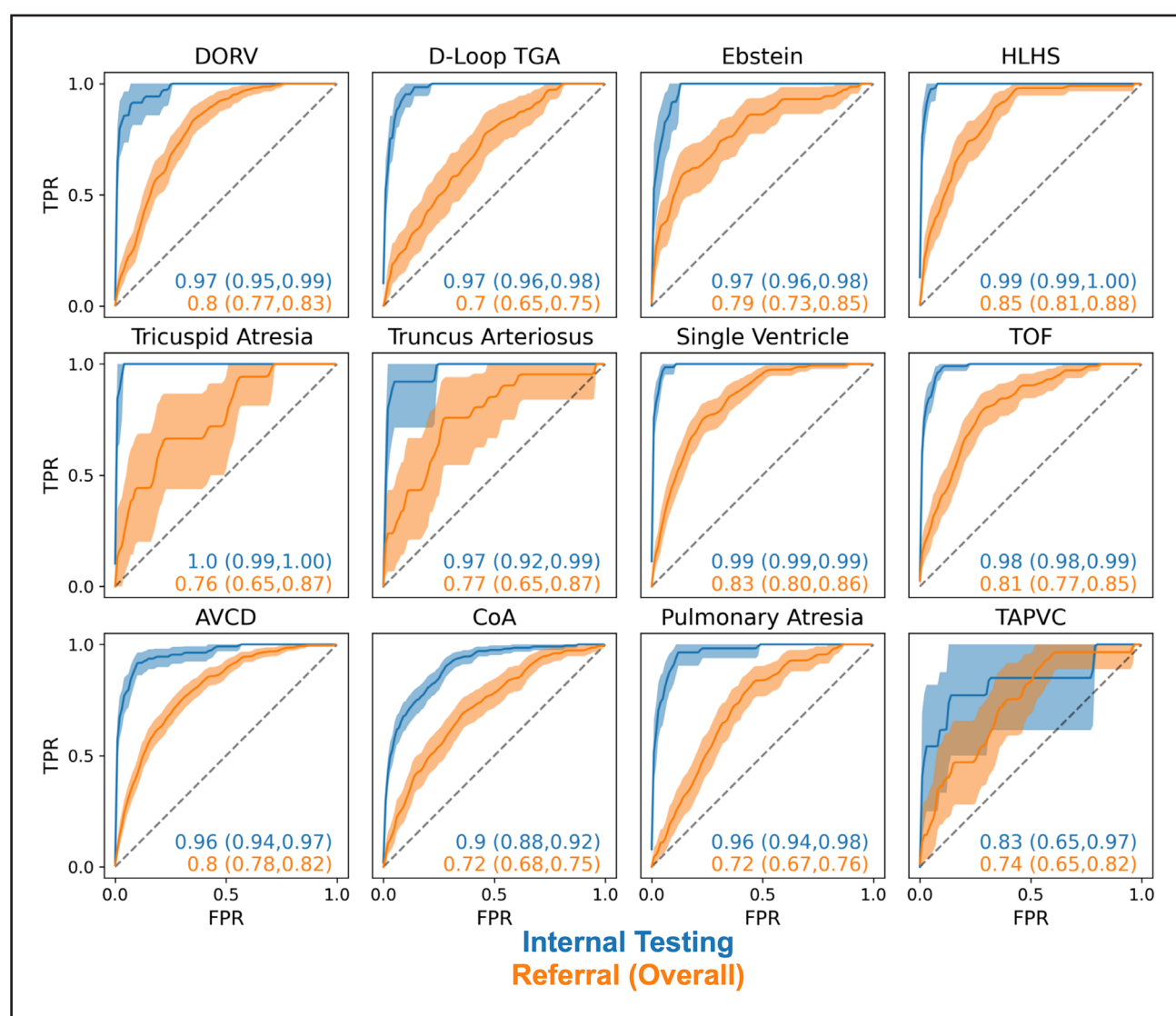


Figure 2. EchoFocus-CHD performance to predict individual critical CHD lesions.

Performance of EchoFocus-CHD to predict individual critical congenital heart disease (CHD) lesions evaluated with the internal (blue) and overall referral (orange) test cohorts using receiver-operating curves. Dotted line represents chance. The 95% CIs are computed with bootstrapping. AVCD indicates atrioventricular canal defect; CoA, coarctation of the aorta; DORV, double-outlet right ventricle; FPR, false-positive rate; HLHS, hypoplastic left heart syndrome; TAPVC, total anomalous pulmonary venous connection; TGA, transposition of the great arteries; TOF, tetralogy of Fallot; and TPR, true-positive rate.

0.72 for ventricular septal defect. [Tables S2 through S5](#) list performance metrics for individual noncritical CHD lesions.

When we assessed the composite critical CHD outcome (Figure 3), internal performance was excellent in both the overall internal cohort (AUROC, 0.94) and the infant subgroup (AUROC, 0.93). In contrast, performance was lower in the referral cohort (AUROC, 0.77 for all referral studies, 0.74 for US referral studies, and 0.82 for international referral studies), which further declined for the infant cohort (AUROC, 0.71 for all referral studies, 0.68 for US referral studies, and 0.73 for international referral studies). Calibration analysis (Figure S1) showed a moderate scaled Brier score

of 0.405 for the internal cohort, whereas the referral cohorts exhibited poor calibration (scaled Brier scores, 0.045 for the overall referral cohort, 0.005 for the US referral cohort, and 0.067 for the international referral cohort).

Subgroup and Sensitivity Analyses

During sensitivity analysis, model performance to detect the composite critical CHD outcome was unchanged when we excluded echos with prior cardiac interventions (internal AUROC, 0.94 [95% CI, 0.93–0.95]; referral cohort AUROC, 0.74 [95% CI, 0.72–0.76]). In addition, using labels generated by a large language model from

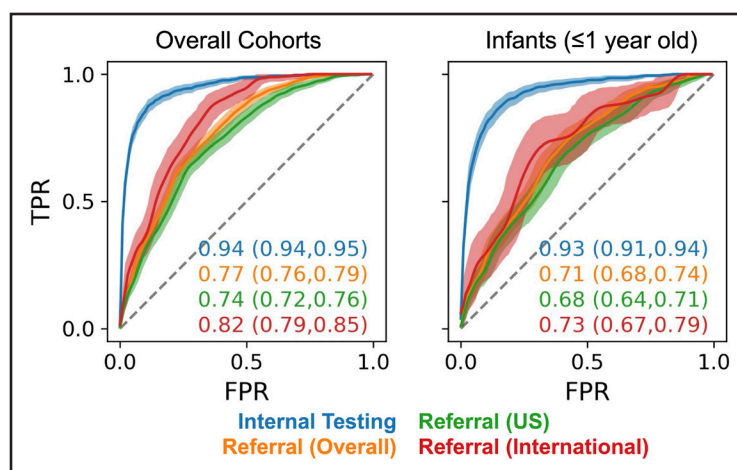


Figure 3. EchoFocus-CHD performance to predict the composite critical CHD outcome.

Performance of EchoFocus-CHD to predict the composite critical congenital heart disease (CHD) outcome evaluated in the overall cohort (left) and infant subgroup (right) with the internal (blue), overall referral (orange), US referral (green), and international referral (red) cohorts using receiver-operating curves. Dotted line represents chance. The 95% CIs are shown with bootstrapping. FPR indicates false positive rate; and TPR, true-positive rate.

echo report free text did not alter model performance (Table S6).

Subgroup analyses by study size demonstrated lower performance for critical CHD detection in studies with <25 videos (Table S7), whereas no consistent performance trends were observed across age subgroups (Table S8).

Expert Adjudication

Expert adjudication was performed on 50 internal and 50 referral cohort discrepant test cases for both false negatives and false positives. For false negatives, adjudicators determined that 42% of internal cases were in fact negative compared with 12% of referral cases (Figure 4A). For false positives, adjudicators determined that 10% of internal cases were indeed positive compared with 20% of referral cases. The distributions of adjudication outcomes differed significantly between the internal and referral cohorts for both false negatives ($P<0.001$) and false positives ($P=0.01$).

Interrater agreement was high internally (Cohen $\kappa=0.82$) with a drop to $\kappa=0.72$ for referral studies, suggesting greater diagnostic ambiguity in the referral cohort. When indeterminate studies were excluded, interrater agreement increased in both cohorts, but the relative decrease in agreement from the internal ($\kappa=0.93$) to referral ($\kappa=0.82$) cohort persisted.

Model Explainability Analysis

Across both internal and referral test cohorts, model explainability on studies with the smallest prediction errors consistently prioritized the same views to detect hypoplastic left heart syndrome and tetralogy of Fallot:

parasternal long-axis, parasternal short-axis, subxiphoid long-axis, and apical views (Figure 5A). In most studies (range, 76%–100%), the top 5 and top 10 attention-weighted clips were sufficient for an expert cardiac imager to determine the presence or absence of critical CHD (Figure 5B). There was no significant difference between internal and referral cohorts in the ability to identify critical CHD from these clips.

Studies with the largest model prediction errors included cases with features mimicking other lesions; for example, a large ventricular septal defect with main pulmonary artery banding was misclassified by EchoFocus-CHD as tetralogy of Fallot (Figure S2).

During view-ablation tests on hypoplastic left heart syndrome and tetralogy of Fallot, the AUROC remained unchanged when any single view (ie, parasternal long-axis, parasternal short-axis, subxiphoid long-axis, or apical view) was removed (Table S9). Notably, performance remained high when restricted to these 4 top views alone (AUROC, 0.98), whereas removal of all 4 reduced performance (AUROC, 0.78–0.93).

Exploring and Addressing Domain Shift

During UMAP visualization, internal studies formed several dense clusters that only partially overlapped with the referral studies (Figure 6A). Figure S3 further visualizes the UMAP embedding, colored by various patient characteristics (eg, age) and CHD types. Notably, some referral studies occupied regions of the embedding space that were sparsely populated by internal studies, suggesting the presence of domain shift. This was particularly evident in the bottom right quadrant, where there was a high density of referral studies with critical CHD (Figure 6B).

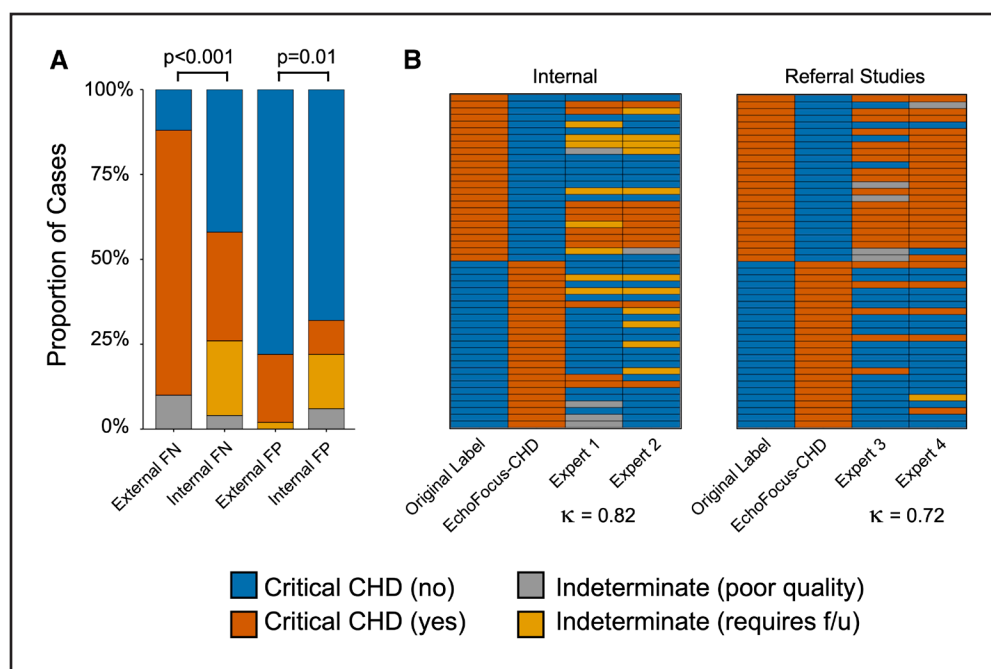


Figure 4. Expert adjudication of discrepant cases.

Expert adjudication was performed on 50 internal and 50 referral discrepant test cases. **A**, Stacked bar plot showing the proportion of cases classified as not critical congenital heart disease (CHD; blue), critical CHD (amber), indeterminate because of poor image quality (gray), and indeterminate because of evolving physiology requiring follow-up (yellow). *P* value obtained with the Fisher exact test. **B**, Heat map displaying study-level classifications assigned by each expert adjudicator; **inset**, interrater agreement assessed with the Cohen κ . FN indicates false negative; and FP, false positive.

Quantitative analyses were subsequently performed to further assess distributional differences between cohorts. We first conducted clustering analysis within the embedding space and observed significant differences in cluster composition by cohort ($\chi^2 P < 0.001$), with referral studies making up the majority of the bottom right cluster (Figure S4). To further evaluate distributional shift, we computed the maximum mean discrepancy between internal and referral embeddings, which demonstrated a statistically significant difference ($P = 0.01$), supporting the presence of cohort-level distributional shift.

To address domain shift, we retrained EchoFocus-CHD using an expanded model development cohort that incorporated all US referral studies in addition to the original BCH training set. The BCH test cohort and the international referral cohort were excluded from training. As shown in Figure 7, internal model performance for the composite critical CHD outcome remained excellent and largely unchanged across the overall cohort, infants, and individual critical CHD lesions (Table S10). Internal calibration was also unchanged (Figure S5). In the international referral cohort, performance for the composite critical CHD outcome improved, with AUROCs of 0.87 for the overall cohort and 0.84 for infants (Figure 7). International referral calibration also improved to 0.151 (Figure S5). For 10 of 12 individual critical CHD lesions, the international referral AUROC increased by a median of 0.08.

Across the retrained internal versus international referral cohorts, composite critical CHD sensitivity was

similar (86%–88%), whereas specificity was lower in the international referral cohort (72% versus 89%). The negative LR was comparable between cohorts (0.13–0.19), whereas the positive LR was higher in the internal cohort (7.8 versus 3.2). Across the internal and international referral cohorts, positive predictive values were 33% (95% CI, 32%–34%) and 35% (95% CI, 30%–41%), respectively; negative predictive values were 99% (95% CI, 99%–99%) and 97% (95% CI, 96%–98%), respectively. Full international referral performance metrics are provided in Table S11.

DISCUSSION

In this study, we developed a view-agnostic, multitask AI-echo model for automated detection of a broad spectrum of CHD lesions. The open-source model introduces a novel study-level transformer encoder as an extension of the PanEcho⁹ framework, enabling integration of information across multiple video clips in a manner analogous to how cardiologists synthesize findings across views, highlighting both the architectural innovation and clinical plausibility of this approach. Using (to the best of our knowledge) the largest pediatric echo dataset reported to date, we demonstrate excellent internal discrimination for both composite and individual CHD outcomes. We further evaluate the generalizability of the model across a large, geographically diverse referral cohort, identifying performance degradation partly attributable to domain shift and

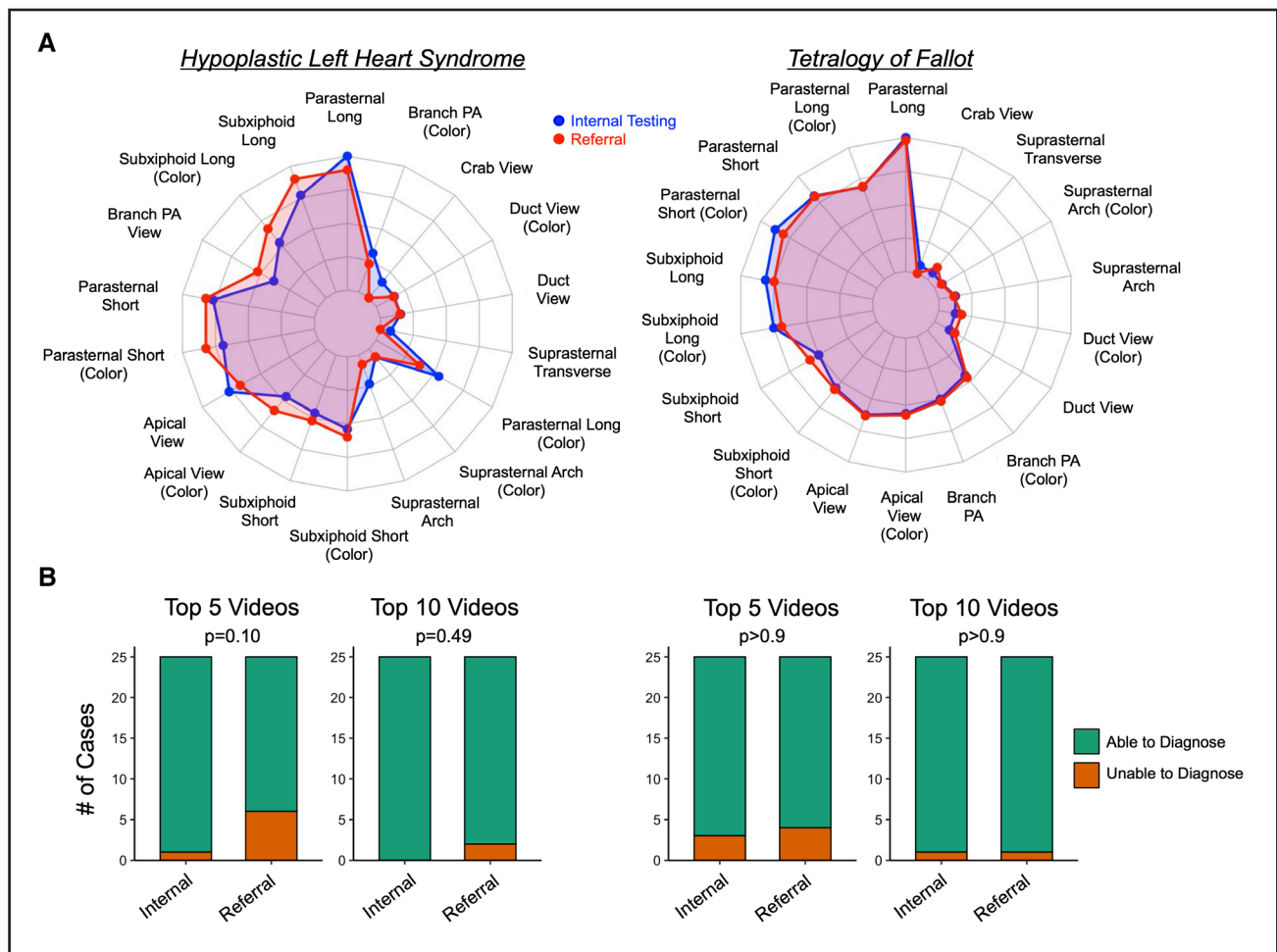


Figure 5. Expert review of EchoFocus-CHD model attention for diagnosing critical CHD.

A, Radar plots of selected views in top 10 clips for hypoplastic left heart syndrome (left) and tetralogy of Fallot (right) for internal (blue) and referral (red) studies. **B**, Diagnostic accuracy of top EchoFocus-CHD selected clips. Stacked bar plots show the proportion of studies in which an expert imager could identify hypoplastic left heart syndrome (left) and tetralogy of Fallot (right) from the top 5 and top 10 clips selected by the model. P values obtained with the Fisher exact test. PA indicates pulmonary artery.

demonstrating that discrimination and calibration can be improved through retraining with more heterogeneous data. Expert adjudication revealed lower interrater referral echo agreement among pediatric cardiologists, suggesting that referral cases missed by the model may represent diagnostically challenging studies rather than unequivocal errors. Altogether, EchoFocus-CHD illustrates the potential of AI-echo to function as a clinical decision-support tool, prioritizing and triaging studies in resource-limited settings to optimize timely access to scarce pediatric cardiology and congenital surgery expertise rather than serving as a replacement for clinician interpretation.

Global Disparities in Pediatric Cardiology Care

There is an underrecognized global burden of pediatric heart disease,²⁸ with CHD constituting a leading cause of childhood noncommunicable mortality worldwide.¹ It is estimated that >90% of children born with CHD reside in low- and middle-income countries, which together

account for 94% of global CHD-related mortality.^{28,29} Even in more developed nations, CHD related mortality is higher in rural and more resource-constrained regions.³⁰ Reducing these inequities is therefore central to achieving the United Nations' Sustainable Development Goals targeting reductions in neonatal mortality and mortality in children <5 years of age by 2030.³¹

Despite this urgency, many pediatric cardiac care systems remain fragile, driven in large part by critical shortages of clinicians with specialized expertise in the diagnosis and management of pediatric heart disease.²⁸ For example, most countries in sub-Saharan Africa and many in Asia lack structured training programs in pediatric cardiology and congenital cardiac surgery³¹ and facilities capable of performing infant or neonatal cardiac surgery.³² Existing models of pediatric heart care in high-income countries are unfeasible for low- and middle-income countries, requiring alternative and context-appropriate strategies to facilitate timely referral to specialized centers. Similar challenges and proposed

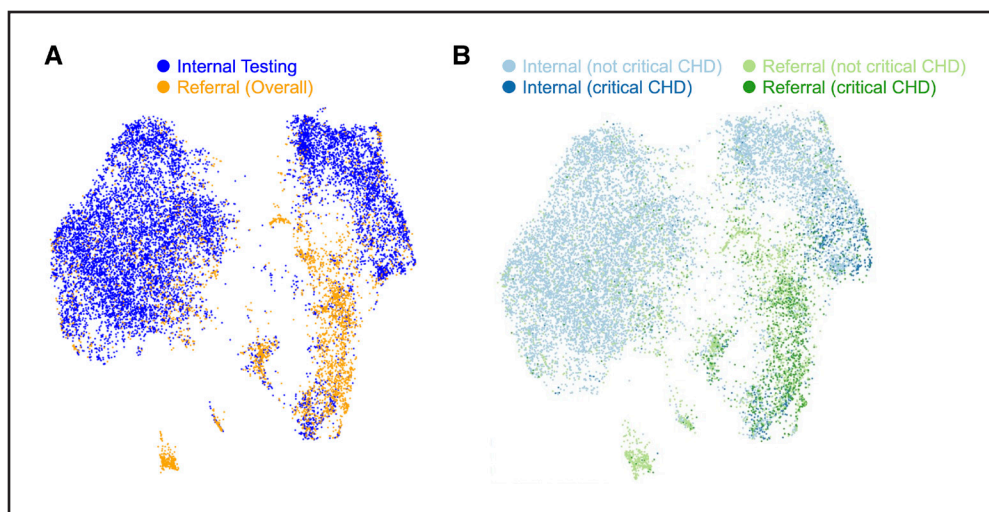


Figure 6. UMAP visualization of study-level embeddings.

A, Visualization of internal (blue) and referral (orange) echo study embeddings projected onto 2 dimensions using uniform manifold approximation and projection (UMAP). **B**, Stratifying internal and referral embeddings by the presence (dark) or absence (light) of critical congenital heart disease (CHD).

solutions have been described in rural and underserved regions of high-income countries such as the United States.

Within this framework, EchoFocus-CHD was developed as an initial step toward enabling scalable, technology-assisted CHD screening and prioritization, with the goal of extending limited pediatric cardiology expertise to settings where it is most constrained.

Clinical Implications of EchoFocus-CHD

EchoFocus-CHD is intended to function as a triage and decision-support tool in resource-constrained settings, where access to pediatric cardiology expertise is limited and timely prioritization of high-risk patients is critical. EchoFocus-CHD may be particularly useful for tele-echocardiography, in which images acquired at community or rural hospitals are transmitted to specialized centers for expert review.³³ Tele-echocardiography programs have shown that remote interpretation can improve diagnostic timeliness, reduce unnecessary transfers, and lower costs.^{33,34} Despite these advances, the global scalability of tele-echocardiography is limited by the small number of pediatric cardiologists available to interpret studies. By potentially reducing reliance on immediate subspecialty interpretation and limiting required views, EchoFocus-CHD may meaningfully enhance the scalability and efficiency of tele-echocardiography infrastructures.

In this context, the operating characteristics of the model support clinically meaningful risk stratification. Internally, EchoFocus-CHD demonstrated high sensitivity and specificity (both $\approx 90\%$), corresponding to strong positive and negative LR (7.8 and 0.13, respectively; Table S10). On referral studies, sensitivity remained high (0.86) with moderate specificity (0.72), yielding a

preserved negative LR of 0.13 and a positive LR of 3.2 (Table S11). These findings indicate that the model is particularly effective for ruling out critical CHD (with negative predictive values of 97%–99% across cohorts), a key requirement for triage applications in which false negatives carry substantial clinical risk. In addition, approximately one-third of cases flagged as positive by EchoFocus-CHD were confirmed to be critical CHD (ie, positive predictive value of 33%–35% across cohorts).

Notably, the performance metrics are similar to a recent AI-enabled echo study that detects a subset of critical CHD lesions³⁵ and likely underestimate true clinical accuracy given that expert adjudication identified a subset of cases initially labeled as incorrect that were correct, evolving physiology (eg, suspected coarctation of the aorta in the setting of a patent ductus arteriosus), diagnostically ambiguous, or challenging even for expert readers. It is important to note that EchoFocus-CHD demonstrated good internal calibration, with improved calibration in the retrained international referral cohort. In low-resource environments where downstream resources such as specialist consultation, transport, or advanced imaging are limited, well-calibrated risk estimates may allow rational prioritization rather than reliance on binary classification alone. Beyond binary triage, EchoFocus-CHD provides lesion-specific subtype predictions, which may further inform urgency, anticipated clinical course, and referral pathways.

Importance of Real-World Deployment

A central objective of this study was to evaluate model performance in a large, diverse cohort that was geographically and demographically distinct from the training population, reflecting conditions expected during

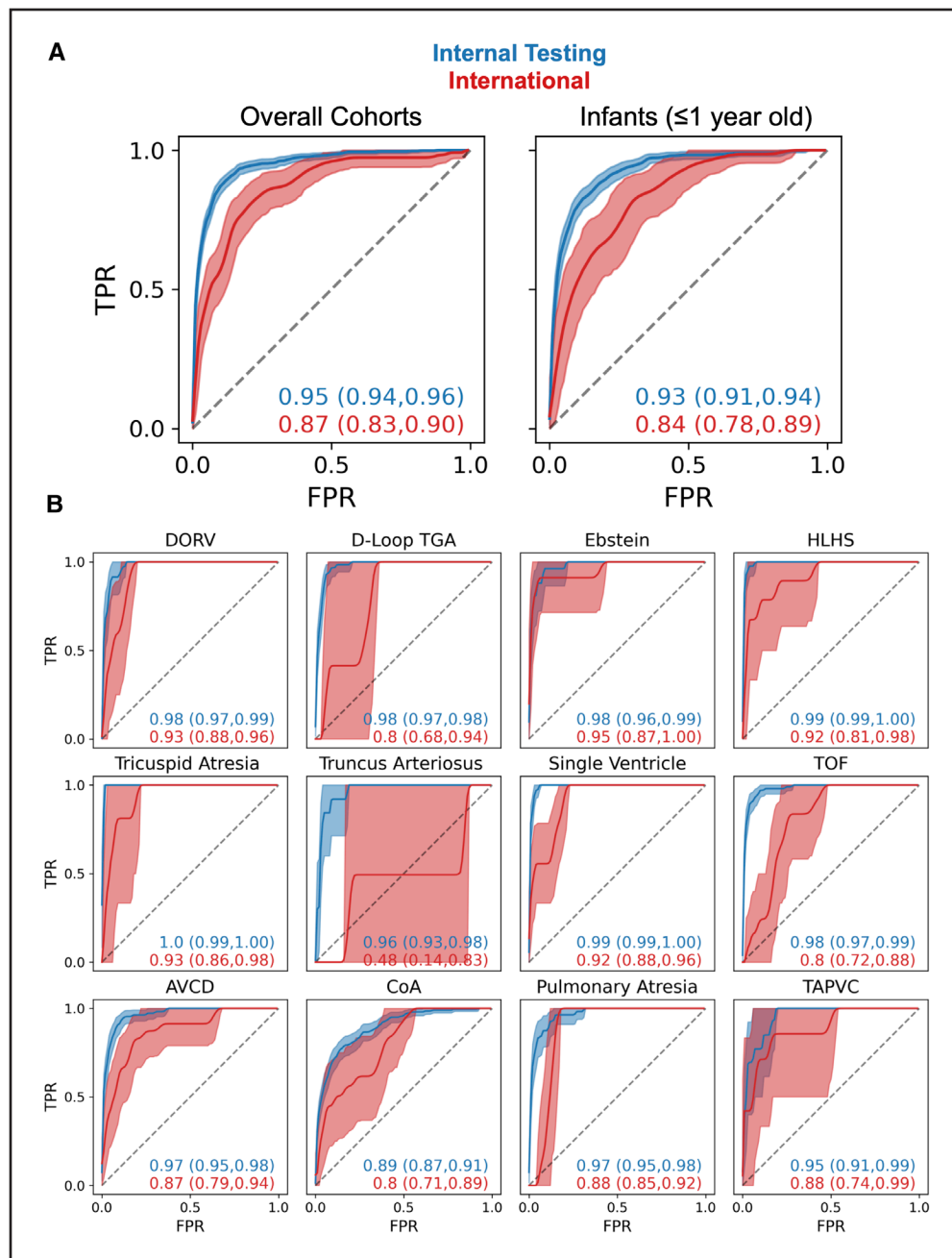


Figure 7. Retraining EchoFocus-CHD on broader dataset improves performance to predict critical CHD.

Performance of retrained EchoFocus-CHD model to predict (A) the composite critical congenital heart disease (CHD) outcome and (B) individual critical CHD outcomes on the internal (blue) and international referral (red) cohorts using receiver-operating curves. Dotted line represents chance. The 95% CIs are computed with bootstrapping. AVCD indicates atrioventricular canal defect; CoA, coarctation of the aorta; DORV, double-outlet right ventricle; FPR, false-positive rate; HLHS, hypoplastic left heart syndrome; TAPVC, total anomalous pulmonary venous connection; TGA, transposition of the great arteries; TOF, tetralogy of Fallot; and TPR, true-positive rate.

real-world deployment. Model performance declined in the referral cohort (Figure 3) independently of outcome labeling approach (ie, Fyler versus large language model), number of videos per study, or patient age. Although the top selected views of the model were consistent across cohorts (Figure 5A) and clinically relevant (Figure 5B), there were discernable differences in the model representations between the internal and referral datasets

(Figure 6). In addition, interrater agreement among expert imagers was lower for referral studies, suggesting greater diagnostic complexity and a lower practical ceiling for discrimination in these cases. Domain shift, whereby the performance of a model degrades when applied to data that differ from its training set, is anticipated in pediatric echo, a modality characterized by substantial variability in vendor-specific image processing, operator-dependent

acquisition techniques, image quality, and institution-specific protocols. These factors introduce meaningful heterogeneity that must be carefully considered as AI-based echo tools move toward clinical deployment.

One way of addressing performance degradation due to domain shift is to incorporate diverse data into the training datasets. The observed improvement in international referral performance after this retraining step highlights the importance of dataset heterogeneity, particularly with respect to pathology diversity and imaging practices, for improving model robustness.

Limitations and Future Directions

Several limitations merit consideration. First, despite retraining on a more heterogeneous US cohort, performance on the international referral cohort remains below the threshold for safe clinical deployment. This highlights the ongoing need to improve model generalizability, which could be addressed through strategies such as (1) exploring alternative/hybrid architectures (eg, EchoPrime)¹⁴ or learning approaches (eg, adversarial learning³⁶); (2) developing a pediatric and CHD-specific foundation model to generate a more robust embedding space; (3) leveraging multi-institutional or federated learning approaches to incorporate data from diverse centers and vendors;³⁷ and (4) multimodal approaches³⁸ such as integrated AI-enabled ECG.^{39–41} Second, although model performance was comparable when either Fyler-coded labels or large language model-extracted labels from echo reports were used, both approaches are imperfect. Nevertheless, the strong agreement between Fyler codes and GPT-4o-mini (Table S12) highlights a potential strategy for multicenter collaboration, enabling automated label generation directly from local echo reports. Third, although our referral validation set was geographically diverse, certain regions of particular clinical interest (most notably sub-Saharan Africa) were not represented, potentially limiting the generalizability of findings to areas with the greatest unmet need. We also acknowledge that the international referral cohort was defined with the use of home address and classification as an outside study rather than the confirmed acquisition site. In addition, the referral nature of the outside studies may potentially introduce selection bias, which may not represent a random sample of community echos for true external validation. Fourth, our models rely on transthoracic echos acquired by trained sonographers; translation to low-resource or point-of-care settings will require further validation on portable ultrasound studies, which may have lower image quality, variable vendors, and greater operator variability. Although encouraging evidence suggests that a reduced set of views may be sufficient (Table S9), robust validation is still required, particularly for extracardiac lesions such as coarctation of the aorta. Fifth, although the model encompasses a broad spectrum of lesions, it does not provide

predictions for all pediatric heart conditions (eg, Kawasaki disease, rheumatic heart disease, cardiomyopathy). Last, although integrated gradients-based explainability was performed, further work is needed to evaluate how these visualizations affect clinician trust and decision-making in practice.

Future directions should include continued model refinement for low-resource settings, prospective multi-site evaluation and external validation in diverse health care environments, and formal assessment of clinical utility and workflow integration.

Conclusions

EchoFocus-CHD demonstrates that large-scale, multi-task AI models show promise to detect a wide range of CHD lesions from routine echo. At the same time, our findings highlight the critical importance of external validation, calibration assessment, and domain shift mitigation for real-world implementation. By identifying both strengths and limitations, this work provides a foundation for future prospective studies and iterative deployment strategies to advance equitable, scalable CHD care worldwide.

ARTICLE INFORMATION

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Disclosures

Dr Mayourian serves as a board member of One Heart Health and as a medical advisor for the Saloni Heart Foundation. Dr Miller on the scientific advisory board for lavita.ai. One Heart Health, the Saloni Heart Foundation, and lavita.ai had no role in the design, conduct, funding, or reporting of this study.

Supplemental Material

Tables S1–S12
Figures S1–S5

REFERENCES

1. Su Z, Zou Z, Hay SI, Liu Y, Li S, Chen H, Naghavi M, Zimmerman MS, Martin GR, Wilner LB, et al. Global, regional, and national time trends in mortality for congenital heart disease, 1990–2019: an age-period-cohort analysis for the Global Burden of Disease 2019 study. *EClinicalMedicine*. 2022;43:101249. doi: 10.1016/j.eclinm.2021.101249
2. van der Linde D, Konings EE, Slager MA, Witsenburg M, Helbing WA, Takkenberg JJ, Roos-Hesselink JW. Birth prevalence of congenital heart

disease worldwide: a systematic review and meta-analysis. *J Am Coll Cardiol*. 2011;58:2241–2247. doi: 10.1016/j.jacc.2011.08.025

3. Oster ME, Lee KA, Honein MA, Riehle-Colarusso T, Shin M, Correa A. Temporal trends in survival among infants with critical congenital heart defects. *Pediatrics*. 2013;131:e1502–e1508. doi: 10.1542/peds.2012-3435
4. Murni IK, Wirawan MT, Patmasari L, Sativa ER, Arafuri N, Nugroho S, Noormanto. Delayed diagnosis in children with congenital heart disease: a mixed-method study. *BMC Pediatr*. 2021;21:191. doi: 10.1186/s12887-021-02667-3
5. Dawson AL, Cassell CH, Riehle-Colarusso T, Grosse SD, Tanner JP, Kirby RS, Watkins SM, Correia JA, Olney RS. Factors associated with late detection of critical congenital heart disease in newborns. *Pediatrics*. 2013;132:e604–e611. doi: 10.1542/peds.2013-1002
6. Zheleva B, Atwood JB. The invisible child: childhood heart disease in global health. *Lancet*. 2017;389:16–18. doi: 10.1016/S0140-6736(16)32185-7
7. Tchervenkov CI, Jacobs JP, Bernier PL, Stellin G, Kurosawa H, Mavroudis C, Jonas RA, Cicek SM, Al-Halees Z, Elliott MJ, et al. The improvement of care for paediatric and congenital cardiac disease across the world: a challenge for the World Society for Pediatric and Congenital Heart Surgery. *Cardiol Young*. 2008;18(suppl 2):63–69. doi: 10.1017/S1047951108002801
8. Aldersley T, Ali S, Dawood A, Edwin F, Jenkins K, Joachim A, Lawrenson J, Reddy D, Boumzebra D, St Louis JD, et al. A landscape analysis of pediatric and congenital heart disease services in Africa. *Cardiol Young*. 2025;35:1782–1791. doi: 10.1017/S1047951125100504
9. Holste G, Oikonomou EK, Tokodi M, Kovacs A, Wang Z, Khara R. Complete AI-enabled echocardiography interpretation with multitask deep learning. *JAMA*. 2025;334:306–318. doi: 10.1001/jama.2025.8731
10. Ouyang D, He B, Ghorbani A, Yuan N, Ebinger J, Langlotz CP, Heidenreich PA, Harrington RA, Liang DH, Ashley EA, et al. Video-based AI for beat-to-beat assessment of cardiac function. *Nature*. 2020;580:252–256. doi: 10.1038/s41586-020-2145-8
11. Sahashi Y, Ieki H, Yuan V, Christensen M, Vukadinovic M, Binder-Rodriguez C, Rhee J, Zou JY, He B, Cheng P, et al. Artificial intelligence automation of echocardiographic measurements. *J Am Coll Cardiol*. 2025;86:964–978. doi: 10.1016/j.jacc.2025.07.053
12. Poterucha TJ, Elias P, Lukyanenko P, Mayourian J. Seeing is believing: intelligence is artificial, responsibility is not. *J Am Coll Cardiol*. 2025;86:979–981. doi: 10.1016/j.jacc.2025.08.029
13. Long A, Haggerty CM, Finer J, Hartzel D, Jing L, Keivani A, Kelsey C, Rocha D, Ruhl J, vanMaanen D, et al. Deep Learning for Echo Analysis, Tracking, and Evaluation of Mitral Regurgitation (DELINEATE-MR). *Circulation*. 2024;150:911–922. doi: 10.1161/CIRCULATIONAHA.124.068996
14. Vukadinovic M, Chiu IM, Tang X, Yuan N, Chen TY, Cheng P, Li D, Cheng S, He B, Ouyang D. Comprehensive echocardiogram evaluation with view primed vision language AI. *Nature*. 2026;650:970–977. doi: 10.1038/s41586-025-09850-x
15. Gearhart A, Goto S, Deo RC, Powell AJ. An automated view classification model for pediatric echocardiography using artificial intelligence. *J Am Soc Echocardiogr*. 2022;35:1238–1246. doi: 10.1016/j.echo.2022.08.009
16. Reddy CD, Lopez L, Ouyang D, Zou JY, He B. Video-based deep learning for automated assessment of left ventricular ejection fraction in pediatric patients. *J Am Soc Echocardiogr*. 2023;36:482–489. doi: 10.1016/j.echo.2023.01.015
17. Reddy C, Yan Y, Qiu M, Tang Y, Jin B, Han Z, Li Y, Zhou S, Tang Q, Xiao H, et al. AI learning for pediatric right ventricular assessment: development and validation across multiple centers. *NPJ Digit Med*. 2025;8:752. doi: 10.1038/s41746-025-02123-x
18. Sharma P, Gearhart A, Luo G, Palepu A, Wang C, Mayourian J, Beam K, Spyropoulos F, Powell AJ, Levy P, et al. Development and validation of a novel deep learning model to predict pharmacologic closure of patent ductus arteriosus in premature infants. *J Am Soc Echocardiogr*. 2025;38:624–632. doi: 10.1016/j.echo.2025.03.018
19. Gearhart A, Anjewierden S, Buddha S, Tandon A. Review of the current state of artificial intelligence in pediatric cardiovascular magnetic resonance imaging. *Children (Basel)*. 2025;12:416. doi: 10.3390/children12040416
20. Kolossvary M, Smith EE, Reeves MJ, Layton A, Mills NL, Motwani M, Goto S, Atkins N. The Transparent Reporting and Evaluation of Artificial Intelligence Models (TRUE-AIM) report card: a quality assurance initiative of the American Heart Association journals to promote high-quality, reproducible artificial intelligence research. *Circulation*. 2025;152:e311–e322. doi: 10.1161/CIRCULATIONAHA.125.074325
21. Colan SD. Early database initiatives: the Fyler codes. In: Barach PR, Jacobs JP, Lipshultz SE, Laussen PC, eds. *Pediatric and Congenital Cardiac Care: Volume 1: Outcomes Analysis*. Springer London; 2015:163–169.
22. Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, Kaiser Ł, Polosukhin I. Attention is all you need. *arXiv*. 2017. doi: 10.48550/arXiv.1706.03762
23. Liu Z, Mao H, Wu C-Y, Feichtenhofer C, Darrell T, Xie S. A ConvNet for the 2020s. *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2022:11966–11976. doi: 10.1109/cvpr52688.2022.01167
24. Loshchilov I, Hutter F. Decoupled weight decay regularization. *arXiv*. 2017. doi: 10.48550/arXiv.1711.05101
25. Srivastava N, Hinton G, Krizhevsky A, Sutskever I, Salakhutdinov R. Dropout: a simple way to prevent neural networks from overfitting. *J Mach Learn Res*. 2014;15:1929–1958.
26. Mayourian J, La Cava WG, Vaid A, Nadkarni GN, Ghelani SJ, Mannix R, Geva T, Dionne A, Alexander ME, Duong SQ, et al. Pediatric ECG-based deep learning to predict left ventricular dysfunction and remodeling. *Circulation*. 2024;149:917–931. doi: 10.1161/CIRCULATIONAHA.123.067750
27. Zhou K, Liu Z, Qiao Y, Xiang T, Change Loy C. Domain generalization: a survey. *arXiv*. 2021. doi: 10.48550/arXiv.2103.02503
28. Kumar RK, Zheleva B, Hasan B. Confronting global inequity in pediatric cardiac care: from crisis to opportunity. *J Am Coll Cardiol*. 2025;86:2161–2163. doi: 10.1016/j.jacc.2025.07.070
29. Deng L, Li Q, Cheng Z. Evaluating the global, regional, and national burden of congenital heart disease in infants younger than 1 year: a 1990–2021 systematic analysis for the GBD study 2021. *Front Pediatr*. 2025;13:1467914. doi: 10.3389/fped.2025.1467914
30. Minhas AMK, Wyand RA, Ariss RW, Nazir S, Jain V, Al-Kindi SG, Shapiro MD, Campbell W, Sperling L, Virani SS. Rural-urban trends in congenital heart disease-related mortality in the United States, 1999 to 2019. *JACC Adv*. 2022;1:100030. doi: 10.1016/j.jaccadv.2022.100030
31. Vervoort D, Jin H, Edwin F, Kumar RK, Malik M, Tapaua N, Verstappen A, Hasan BS. Global access to comprehensive care for paediatric and congenital heart disease. *CJC Pediatr Congenit Heart Dis*. 2023;2:453–463. doi: 10.1016/j.cjpc.2023.10.001
32. Awuah WA, Adebuseye FT, Wellington J, Ghosh S, Tenkorang PO, Machai PNM, Abdul-Rahman T, Mani S, Salam A, Papadakis M. A reflection of Africa's cardiac surgery capacity to manage congenital heart defects: a perspective. *Ann Med Surg (Lond)*. 2023;85:4174–4181. doi: 10.1097/MS9.0000000000001054
33. Satou GM, Rheuban K, Alverson D, Lewin M, Mahnke C, Marcin J, Martin GR, Mazur LS, Sahn DJ, Shah S, et al; American Heart Association Congenital Cardiac Disease Committee of the Council on Cardiovascular Disease in the Young and Council on Quality Care and Outcomes Research. Telemedicine in pediatric cardiology: a scientific statement from the American Heart Association. *Circulation*. 2017;135:e648–e678. doi: 10.1161/CIR.0000000000000478
34. Randolph GR, Hagler DJ, Khandheria BK, Lunn ER, Cook WJ, Seward JB, O'Leary PW. Remote telemedical interpretation of neonatal echocardiograms: impact on clinical management in a primary care setting. *J Am Coll Cardiol*. 1999;34:241–245. doi: 10.1016/s0735-1097(99)00182-5
35. Joseph C, Mrudang M, Dhamanpreet K, Matthew D, Adil D, Aravind K, Matthew L, Rohan S, Gonzalez AK, Joseph L, et al. A multi-task deep learning model for pediatric echocardiography analysis. *medRxiv*. Preprint posted online October 30, 2025. doi: 10.1101/2025.10.27.25338912
36. Donahue J, Krähenbühl P, Darrell T. Adversarial feature learning. 2016. *arXiv*. doi: 10.48550/arXiv.1605.09782
37. Goto S, Solanki D, John JE, Yagi R, Homilius M, Ichihara G, Katsumata Y, Gaggin HK, Itabashi Y, MacRae CA, et al. Multinational federated learning approach to train ECG and echocardiogram models for hypertrophic cardiomyopathy detection. *Circulation*. 2022;146:755–769. doi: 10.1161/CIRCULATIONAHA.121.058696
38. Lai C, Yin M, Kholmovski EG, Popescu DM, Lu DY, Scherer E, Binka E, Zimmerman SL, Chrispin J, Hays AG, et al. Multimodal AI to forecast arrhythmic death in hypertrophic cardiomyopathy. *Nat Cardiovasc Res*. 2025;4:891–903. doi: 10.1038/s44161-025-00679-1
39. Mayourian J, O'Sullivan D, Bravo-Jaimes K, Madhavan M, Ghelani S, Geggel R, Triedman JK. Infant electrocardiogram-based deep learning predicts critical congenital heart disease. *JACC Clin Electrophysiol*. 2025;11:2757–2759. doi: 10.1016/j.jacep.2025.08.005
40. Ghelani SJ, Thattai N, La Cava W, Triedman JK, Mayourian J. Artificial intelligence-enabled ECG to detect congenitally corrected transposition of the great arteries. *Pediatr Cardiol*. 2026;47:1376–1382. doi: 10.1007/s00246-025-03916-3
41. Mayourian J, Geggel R, La Cava WG, Ghelani SJ, Triedman JK. Pediatric electrocardiogram-based deep learning to predict secundum atrial septal defects. *Pediatr Cardiol*. 2025;46:1235–1240. doi: 10.1007/s00246-024-03540-7